

Lob-4

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CS24B055

Task-I

$$F(A, B, C) = A \cdot \bar{B} + BC$$

Truth table $\Rightarrow$

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Task-IV

$Q_1 = \text{MSB}$

$Q_0 = \text{LSB}$

For Up Counter,

<u>D_1</u>	<u>D_0</u>	<u>Q_1</u>	<u>Q_0</u>
0	1	0	0
1	0	0	1
1	1	1	0
1	1	1	1

Karnaugh map

For D_1

Q_1	Q_0
0	0
0	1
1	0
1	1

$$\Rightarrow D_1 = Q_0 + Q_1$$

For D_0

Q_1	Q_0
0	0
0	1
1	0
1	1

$$\Rightarrow D_0 = \bar{Q}_0 + Q_1$$

For Down Counter,

<u>D_1</u>	<u>D_0</u>	<u>Q_1</u>	<u>Q_0</u>
1	0	1	1
0	1	1	0
0	0	0	1
0	0	0	0

Karnaugh map

For D_1

Q_1	Q_0
0	0
0	1
1	0
1	1

$$\Rightarrow D_1 = Q_1 \cdot Q_0$$

Similarly, for D_0

$$\Rightarrow D_0 = Q_1 \cdot \bar{Q}_0$$